

Lab 12: Practice Lab for Programming

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Instructions: This lab is meant for your practice, You need not submit anything for this lab.

Question 1: Using Loops

1. Write a Python program that prints the following pattern of numbers based on the size of an input. The program should take a positive integer n (e.g., 5) as input, and print a triangle-like structure with even and odd numbers labeled differently.

Instructions: - Use ****nested loops**** to generate the pattern. - Use ****if-else**** conditions to print "Even" or "Odd" for each number.

Sample Input:

```
n = 5
```

Expected Output:

```
1: Odd
2: Even 3: Odd
4: Even 5: Odd 6: Even
7: Odd 8: Even 9: Odd 10: Even
11: Odd 12: Even 13: Odd 14: Even 15: Odd
```

2. Simulate a guessing game using a **while** loop. The program selects a random number between 1 and 100, and the user has to guess the number. For each incorrect guess, print whether the guess was too high or too low. Use **break** to exit the loop when the correct guess is made.

Sample Input:

```
Guess the number: 50
Too high!
Guess the number: 30
Too low!
Guess the number: 42
Congratulations! You guessed it!
```

3. A palindrome number is a number that reads the same forward and backward. For example, 525, 1441, and 123321 are palindrome numbers.
 - Write a program to check if a number is palindrome or not.
 - Write a program to check if a number is prime palindrome.

Question 2: Game of Survival

In a unique game, n participants stand in a circle, each assigned a position from 1 to n in sequential order. The game is organized such that the game master eliminates every alternate participant, continuing this pattern around the circle until only one person remains.

The rules are as follows:

- Starting with the first person in the circle, the game master eliminates them.

- The next person (the one immediately clockwise) is skipped and remains in the circle.
- The game master then moves to the next person after the one just skipped and eliminates them.
- This pattern continues: each alternate person in the circle is eliminated, while the others remain. The circle closes in as people are eliminated.
- The process repeats until only one person remains in the circle.

The last person remaining in the circle is declared the winner of the game.

1. For a given number of participants n , simulate this elimination process by coding it step-by-step to determine the winner's position.
2. Then, based on your observations, try to derive a general formula to determine the position of the winner for any n participants.

Hint: The expected answer for $n=7$, is 6; for $n=13$ is 10, $n=16$ is 16

3. Consider a variant where elimination starts from the second position. How does your solution changes

Question 3. Policy Design: Determining Pricing ¹

Note: Make sure the dimensional consistency and unit consistency are maintained for objective and constraint formulations.

Suppose you are employed as an auditor by the government to a leading Oil and Gas Company. As an auditor, you are bound by the government to design policies that the company needs to follow. The product portfolio for the company includes three finished products available for consumption by consumers as P_1 , P_2 , and P_3 . The company acquires some raw material and converts it into by-products B_1 and B_2 , which further are processed to produce P_1 , P_2 , P_3^A and P_3^B . It is estimated that 600000 and 750000 (in tonnes) of B_1 and B_2 will be available for finished production in the coming year. Table 1 below provides the content of B_1 and B_2 (in %) in finished products.

	B_1	B_2
P_1	4 %	9 %
P_2	80 %	2 %
P_3^A	35 %	30 %
P_3^B	25 %	40 %

Table 1: Percentage of content of B_1 and B_2 in finished product

As an auditor, you are also responsible for cost accounting and price determination based on market situation. From experience, it is known that price and demand are relatively sensitive, and Table 2 below provides a relationship between price and demand as the coefficient of elasticity.

P_1	P_2	P_3^A	P_3^B	P_3^A to P_3^B	P_3^B to P_3^A
0.4	2.7	1.1	0.4	0.1	0.4

Table 2: Price (cross) elasticity to demand for finished products

Herein, the market study suggests that prices and demand are sensitive to each other. Suppose e represents the elasticity of demand for a product, i.e., how sensitive your demand is against the price of that

¹Louwes, Stephanus Louwe, John CG Boot, and S. Wage. "A Quadratic-programming Approach to the Problem of the Optimal Use of Milk in the Netherlands." Journal of Farm Economics 45.2 (1963): 309-317.

particular product (Δ change in demand *w.r.t* Δ change in price). Further, e_{ij} denotes the cross elasticity of product i to product j wherein the demand of product i impacts the price of product j (Δ change in demand for i *w.r.t* Δ change in price of j).

	Consumption (in 1000 tons)	Price (in \$/ton)
P_1	4820	297
P_2	320	720
P_3^A	210	1050
P_3^B	70	815

Table 3: Price and Demand of last year for finished products

From the previous year's data, consumption and price for each product are given in Table 3. Capacity constraints for the production of finished products from by-products restrict production of finished products. Also, to ensure a competitive advantage in the market, it is highly undesirable to allow certain price index to rise. Therefore, this limitation simply demands that the new prices must be such that the total cost of last year's revenue from consumption would not be increased. Your objective is to determine the optimal price of the product portfolio along with the production of finished products for the company under market conditions, given that capacity constraints are satisfied for maximum revenue.

Hint:

Use the following relationship(s):

$$\frac{dx_i}{x_i} = -e_i \frac{dp_i}{p_i} + e_{ij} \frac{dp_{ij}}{p_{ij}}$$

is a differential equation representing the relationship between price and elasticity. **Note:** To use the above expressions in constraints and objective functions will result in non-linearities. You can convert the above differential equation into difference equation as:

$$\frac{x_i - \bar{x}_i}{\bar{x}_i} = -e_i \frac{p_i - \bar{p}_i}{\bar{p}_i} + e_{ij} \frac{p_{ij} - \bar{p}_{ij}}{\bar{p}_{ij}}$$

1. Write your optimization formulation from the above description to determine the optimal price for the product portfolio. Describe your formulation (constraints, objective functions, and support for decision variables) and assumption along with the necessary assumption used.
2. Your formulation by nature involved a quadratic objective resulting quadratic optimization problem (due to price and quantity both being decision variables). Suggest how you will modify the above formulation to reduce your complexity. (**Hint:** Try to difference equation!!).
3. Suggest how you can check the nature of P for your formulation and comment on observation about the above formulation based on nature of P .
4. Model your formulation as a Quadratic Optimization Problem and solve it using *qpsolvers* or *gurobi* solver or any other appropriate approach. Report optimal pricing for product portfolio.
5. Imagine that change in disposable income can vary -10 % to 5 % from the previous year to the new year, thus leading to a rise/fall in inflation as consumers are willing to pay more/less. Plot a curve to show how sensitive your revenue from consumption is against changes in disposable income. Comment on the observations made from the plot.

Question 4:

Alice was given the n integers from 1 to n . She wrote all possible permutations in increasing lexicographical order, and wrote each permutation in a new line. For example, for $n = 3$, there are 6 possible permutations:

1. 1,2,3
2. 1,3,2
3. 2,1,3
4. 2,3,1
5. 3,1,2
6. 3,2,1

She then chose one permutation among them as her favorite permutation. After some time, she forgot some elements of her favorite permutation. Nevertheless, she still tried to write down its elements. She wrote a 0 in every position where she forgot the true value. She wants to know the sum of the line numbers of the permutations which could possibly be her favorite permutation, i.e., permutations which can be obtained by replacing the 0s. Can you help her out?

Sample Input:

```
4          #value of n
0 2 3 0    #space separated permutation
```

Sample Output:

```
23
```

The possible permutations are [1,2,3,4] and [4,3,2,1]. The permutation [1,2,3,4] occurs on line 1 and the permutation [4,3,2,1] occurs on line 22. Therefore the sum is $22+1=23$.